

Task 1

Understanding Dataset & Data Types

Tools Used: Python, Pandas, NumPy, Jupyter Notebook

Introduction

This project focuses on understanding dataset structure and data types using Python libraries such as Pandas and NumPy.

The analysis includes identifying numerical and categorical variables, checking missing values, exploring dataset information, and generating statistical summaries.

This task helped build a foundation in Exploratory Data Analysis (EDA).

Objective

- Load datasets using Pandas
- Understand dataset structure
- Identify numerical and categorical variables
- Detect missing values
- Generate statistical summaries

Tools & Technologies Used

- Python
- Pandas
- NumPy
- Jupyter Notebook

Dataset Used

Dataset Name: dataset.csv

The dataset contains multiple columns with different data types including integers, floating-point numbers, and categorical values.

Steps Performed

1. Imported required libraries
2. Loaded the dataset
3. Displayed first 5 rows
4. Checked dataset shape
5. Used `df.info()` to inspect data types
6. Used `df.describe()` for statistical summary
7. Checked missing values
8. Found unique values in categorical columns

Important Code Snippets

```
import pandas as pd
import numpy as np
```

```
df = pd.read_csv('dataset.csv')
df.head()
df.info()
df.describe()
df.isnull().sum()
```

Understanding Data Types

Numerical Data:

- Integer (int)
- Float (float)

Categorical Data:

- Labels or categories stored as object data types

Continuous vs Discrete Variables

Continuous Variables:

- Height
- Weight

Discrete Variables:

- Number of students
- Number of cars

Handling Missing Values

Common methods:

- `df.dropna()`
- `df.fillna(0)`

Key Observations

- Dataset contains both numerical and categorical data
- Some columns may contain missing values
- Statistical summaries help understand numerical columns

Learning Outcome

- Learned dataset exploration using Pandas
- Understood different data types
- Learned basic Exploratory Data Analysis (EDA)

Conclusion

This task provided a strong foundation in understanding datasets and performing basic exploratory data analysis using Python.